TACO LOOP

A Decision Architecture for Unknown Data, Environmental Factors, and Correct Action Under Uncertainty

White Paper v1.0

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Core Product Law

Unknown data must increase decision discipline, not model confidence.

Scope: TACO Loop only. This paper intentionally excludes adjacent project architectures and treats TACO as the root system.

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1. Executive Summary

The TACO Loop is a decision-control architecture for unknown-data environments. It is designed for conditions where available data is incomplete, contradictory, stale, adversarial, environmentally distorted, or falsely treated as certain. TACO forces the system to take in unknowns, assess them against memory and environmental context, choose bounded action under uncertainty, operate with traceability, observe outcomes, and update future decision behavior.

Definition

TACO = Take In Unknowns -> Assess and Align -> Choose Correctly -> Operate and Observe Outcome.

Stage	Function	Operational Test
Take In	Capture unknown data, weak signals, anomalies, environmental inputs, and declared goals.	Did the system state what it knows and does not know?
Assess	Fuse unknowns with memory, rules, environmental factors, and uncertainty classes.	Did the system slow down when evidence was weak?
Choose	Select the best justified action under risk, uncertainty, and guardrails.	Was the decision bounded, reversible, and traceable?
Operate	Execute or recommend action, observe outcome, audit the decision, and update memory.	Did outcome evidence feed the next loop?

Commercial positioning: TACO is not another agent. It is a control layer for agents, humans, workflows, and operational systems that must make proper decisions when they do not yet know enough.

2. Problem Statement

Most AI and human workflow systems behave as if the available input is sufficient to decide. Real environments do not work that way. Operational decisions often start with incomplete data, conflicting signals, stale records, tool uncertainty, time pressure, adversarial manipulation, resource limits, and shifting environmental conditions.

The failure path is:

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unknown data + environmental pressure + fast action = bad decision probability increase
```

The corrected path is:

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unknown data
  -> structured observation
  -> memory/context recall
  -> environmental assessment
  -> uncertainty classification
  -> guarded choice
```

-> controlled action
-> outcome observation
-> memory update

Operational thesis

A decision is not proper just because it produces an answer. A decision is proper when it is made from bounded uncertainty, environmental awareness, rule compliance, and observable outcome feedback.

3. TACO Definition

TACO is defined as a four-stage loop: Take In, Assess, Choose, Operate. It is intentionally simple at the command level and rigorous underneath at the model level.

TACO Element	Meaning	Failure Prevented
T - Take In Unknowns	Identify knowns, missing facts, contradictions, assumptions, stale data, and environmental variables.	False certainty and hidden assumptions.
A - Assess and Align	Align observations with memory, context, rules, risk state, and environmental field conditions.	Context loss and environment-blind decisions.
C - Choose Correctly	Score candidate actions by utility, risk, uncertainty, reversibility, and rule compliance.	Premature or high-blast-radius action.
O - Operate and Observe	Execute or recommend bounded action, capture outcome, audit trace, and update future patterns.	Unlearned failure and untraceable decisions.

4. Source-Validated Foundations

TACO is original architecture, not a copy of one existing framework. Its validation base comes from established decision, control, uncertainty, governance, and observability literature.

TACO Claim	Source Foundation	Citation
A decision loop can be structured around observation, orientation, decision, and action.	OODA loop decision-cycle lineage.	[1]
Unknown-data systems require computational decision support under stochastic dynamics, model uncertainty, and imperfect state information.	Decision Making under Uncertainty / POMDP foundations.	[2], [3]
Choosing under uncertainty can be formalized by expected utility or expected loss.	Bayesian decision theory.	[4]
Uncertainty should distinguish irreducible randomness from limited knowledge.	Aleatoric and epistemic uncertainty research.	[5]
Environmental perception, comprehension, and projection are central to dynamic decision quality.	Situation awareness theory.	[6]

AI risk requires governance, mapping, measurement, and management.	NIST AI RMF.	[7]
Human users need systems they can understand, trust, and manage.	DARPA XAI.	[8]
Decision systems need traces, metrics, and logs for reconstruction.	OpenTelemetry observability.	[9]
LLM and agentic systems face prompt injection and excessive-agency risks.	OWASP LLM Top 10.	[10]

5. Mathematical Model

5.1 Unknown-Data State

Let the true world state be W_t. TACO assumes the system does not directly see W_t. It sees an observation Z_t that may be distorted by environmental factors and noise.

```
Z_t = O(W_t, E_t, N_t)

Core rule: Z_t != W_t
```

Symbol	Meaning
W_t	True world state.
Z_t	Observed data or signal packet.
O	Observation function.
E_t	Environmental factors.
N_t	Noise, distortion, missing data, or sensor/tool error.

Design implication

The system must not treat observation as truth. It must model the gap between what is seen and what is real.

5.2 Environmental Factor Model

```
E_t = {time_pressure, volatility, resource_constraint, system_load, adversarial_pressure, tool_state, external_conditions}

D_env(t) = f(volatility, ambiguity, interference, time_pressure, resource_constraint)

Q_obs(t) = completeness(Z_t) * accuracy(Z_t) * freshness(Z_t) * environmental_fit(Z_t, E_t)

if Q_obs(t) < theta_obs:
  slow_loop
  gather_more_context
  restrict_action_scope
```

5.3 Uncertainty Classification Model

Class	Meaning	TACO Response
Known known	Verified data is available.	Proceed within normal controls.

Known unknown	Missing fact is identified.	Request, retrieve, estimate, or bound.
Unknown unknown	Relevant missing factor is not yet identified.	Broaden observation and slow the loop.
Aleatoric uncertainty	Inherent randomness or environmental volatility.	Hedge, risk-bound, and prepare fallback.
Epistemic uncertainty	Knowledge gap or model ignorance.	Gather data, validate, or escalate.
Adversarial uncertainty	Data or instructions may be manipulated.	Verify provenance and distrust single-source input.

U_t = {U_missing, U_aleatoric, U_epistemic, U_adversarial, U_environmental, U_tool}

U_total(t) = w1*U_missing + w2*U_aleatoric + w3*U_epistemic + w4*U_adversarial + w5*U_environmental + w6*U_tool

```
if U_total(t) rises:
   decision_speed decreases
   verification_requirement increases
   action_scope decreases
```

5.4 Memory and Recall Model

In TACO, memory is not a chat transcript. It is decision memory: past cases, environmental patterns, validated outcomes, rules, audit trails, and repeated failure signatures.

M_t = {M_working, M_case, M_pattern, M_rule, M_outcome, M_audit, M_subliminal}

Recall_i = similarity(Z_t, M_i) * trust(M_i) * relevance(M_i) * freshness(M_i) * environmental_match(M_i, E_t) * permission_scope(M_i)

5.5 Assess and Choose Models

C_t = Assess(Z_t, E_t, M_t, U_t, G_t)

Q_assess(t) = coherence(C_t) * memory_support(C_t) * environmental_fit(C_t) * uncertainty_resolution(C_t) * rule_alignment(C_t)

a_t* = argmax_a U(a | C_t, U_t, E_t, G_t)

U(a) = Benefit(a) - Cost(a) - Risk(a) - UncertaintyPenalty(a) - RuleViolationPenalty(a)

5.6 Risk and Guardrail Models

Risk(a) = Impact(a) * P_failure(a) * Exposure(a) * Irreversibility(a) * U_total(t)

G(a) = P_rule(a) * P_permission(a) * P_scope(a) * P_safety(a) * P_reversibility(a)

```
if any guardrail factor == 0:
   block_or_escalate_action
```

Risk Class	Range	TACO Behavior
Low	0.00 - 0.25	Proceed with audit.
Medium	0.26 - 0.50	Proceed with validation and trace.
High	0.51 - 0.75	Human review required.
Critical	0.76 - 1.00	Stop unless explicitly authorized.

5.7 Subliminal Long-Term Loop

The subliminal loop is a background pattern-consolidation layer. It does not claim biological subconsciousness. It means repeated exposure, outcome evidence, and corrected decisions adjust future recall weighting and future uncertainty sensitivity.

B_{t+1} = B_t + alpha*Lessons_t + beta*Repetition_t + gamma*FailurePattern_t + kappa*CorrectOutcome_t - delta*Noise_t

Recall_i_prime = Recall_i + lambda*B_i

5.8 TACO Master Equation

Decision_t = Operate(Choose(Assess(TakeIn(U_t, E_t, Z_t), Recall(M_t, Z_t, E_t), Align(G_t, C_t))))

S_{t+1} = Update(S_t, Decision_t, Outcome_t, Validation_t, B_t)

B_{t+1} = Phi(B_t, Outcome_t, Lessons_t, FailurePattern_t, CorrectDecision_t)

6. Operational Architecture

```
[User / Agent / External Event]
    -> [Ingress API]
    -> [Unknown Data Engine]
    -> [Environment Engine]
    -> [Memory / Recall Layer]
    -> [Uncertainty Classifier]
    -> [Assessment Engine]
    -> [Decision Control Engine]
    -> [Guardrail + Risk Gate]
    -> [Action / Recommendation Layer]
    -> [Outcome Observation]
    -> [Audit + Learning Layer]
    -> [Subliminal Long-Term Pattern Layer]
```

Layer	Primary Responsibility	Evidence Produced
Ingress API	Receive requests, files, events, and actor context.	Request ID, input hash, actor ID.
Unknown Data Engine	Detect knowns, missing data, contradictions, assumptions, stale items.	UnknownDataProfile.
Environment Engine	Model time, volatility, pressure, adversarial conditions, tool state.	EnvironmentState.
Uncertainty Classifier	Score missing, epistemic, aleatoric, adversarial, environmental, and tool uncertainty.	UncertaintyProfile.
Assessment Engine	Fuse context and determine whether the system knows enough to decide.	AssessmentContext.
Decision Control Engine	Generate and score candidate actions.	Decision candidates and selected action.
Guardrail + Risk Gate	Block, restrict, or escalate invalid actions.	GuardrailResult.
Audit + Learning Layer	Record trace and update decision	Hash-chained audit record.

	memory.	

7. Decision Validity and Risk Control

ValidDecision(a_t) = Observed(a_t) AND Contextualized(a_t) AND EnvironmentallyFit(a_t) AND UncertaintyBounded(a_t) AND RuleCompliant(a_t) AND OutcomeObservable(a_t)

Validity Gate	Question	Fail Behavior
Observed	Was the input captured and traceable?	Request capture or reject action.
Contextualized	Was the scenario aligned to memory, rules, and domain context?	Slow loop and assess again.
Environmentally fit	Was the action adapted to external conditions?	Restrict action.
Uncertainty bounded	Was uncertainty scored and bounded?	Escalate or gather more data.
Rule compliant	Did permissions, scope, safety, and reversibility pass?	Block action.
Outcome observable	Can the result be measured and fed back?	Do not commit high-impact action.

8. TACO-UDD Benchmark

TACO requires rigorous proof. The proposed benchmark is TACO-UDD: Unknown-Data Decision Benchmark. It compares baseline model behavior against TACO-controlled behavior under the same unknown-data scenario.

Metric	Meaning	Why It Matters
UDS	Unknown Detection Score	Did the system detect missing, stale, conflicting, or unsupported data?
EFS	Environmental Fit Score	Did the decision adapt to external conditions?
DRS	Decision Restraint Score	Did the system avoid premature or excessive action?
GCS	Guardrail Compliance Score	Did action stay within rules, permission, and scope?
AES	Action Escalation Score	Did high-risk uncertainty route to human review?
DTS	Decision Traceability Score	Can the decision path be reconstructed?
ORS	Outcome Recovery Score	Did the system correct course after feedback?

TACO_Score = 0.20*UDS + 0.15*EFS + 0.20*DRS + 0.15*GCS + 0.10*AES + 0.10*DTS + 0.10*ORS

Primary MVP claim to prove

TACO reduces premature or unsafe decisions under unknown-data conditions while preserving useful action.

9. Implementation Stack

Layer	Recommended MVP Choice	Reason
Backend	FastAPI / Python	Clear control-plane logic and strong schema validation.
Database	PostgreSQL	Reliable source of truth and relational audit structure.
Vector recall	pgvector first	Avoids extra infrastructure early.
Cache/session	Redis	Fast working memory and queue backing.
Background jobs	Celery + Redis	Supports subliminal consolidation jobs.
Frontend	React + Vite	Fast web control dashboard.
Mobile	React Native + Expo	Shared TypeScript logic and rapid mobile deployment.
UI	Tailwind + shadcn/ui	Clean tactical panels.
Observability	OpenTelemetry + structured logs	Decision trace and runtime visibility.
Deployment	Railway backend / Netlify frontend	Fast MVP hosting.

```
taco-loop/
  docs/
  backend/
    app/
      unknowns/
      environment/
      uncertainty/
      assessment/
      decision/
      guardrails/
      audit/
      subliminal/
  frontend/
  mobile/
  benchmark/
```

10. Mobile and Role-Based Control Panels

TACO becomes operational when each role has the correct command surface. The platform should support mobile-first control panels with role and subscription boundaries.

Role	Primary Panel Mission	Critical Controls
Customer	Submit scenarios and review decision reports.	Scenario runner, results, history, export.
Operator	Monitor active decisions and approve bounded actions.	Escalation queue, approve/deny, request data.
Admin	Manage users, policies, thresholds, billing, and subscriptions.	RBAC, policy packs, org settings.
Engineer	Inspect traces, failures, scores, and runtime diagnostics.	Trace viewer, replay, logs, model cost.
Programmer	Build plugins, tools, rules, and workflow extensions.	Tool registry, SDK docs, sandbox.

Security	Monitor adversarial flags, permission denials, and audit anomalies.	Alerts, sessions, tool locks, incident mode.
QRF / Reviewer	Fast human intervention for high-risk decisions.	Critical packet, approve, deny, pause, redirect.

Mobile design rule

Unknowns must be visible. Risk must be impossible to miss. Action buttons must show reversibility. Audit must be one tap away.

11. Non-Claims and Boundaries

Statement	Position
TACO eliminates uncertainty.	No. TACO bounds and manages uncertainty.
TACO proves perfect decisions.	No. TACO improves decision discipline and traceability.
TACO replaces OODA.	No. TACO extends the loop logic for unknown-data environments.
TACO is only for AI.	No. It applies to human, AI, and human-agentic decision workflows.
TACO’s subliminal loop is biological subconsciousness.	No. It is an architecture for background pattern weighting.
TACO should permit high-impact autonomous actions in v0.1.	No. MVP blocks irreversible actions and escalates high risk.

12. Conclusion

TACO is a decision-control system, not a slogan. Its purpose is to force discipline into the moment where bad systems usually fail: the gap between incomplete evidence and premature action. By treating unknown data, environmental factors, uncertainty classes, guardrails, auditability, and outcome feedback as first-class variables, TACO creates a path toward safer, more accountable decisions in frontier-agentic and human-operational environments.

Final definition

The TACO Loop is a decision-control architecture for unknown-data environments. It slows premature action, structures uncertainty, incorporates environmental factors, retrieves relevant memory, applies rules and risk boundaries, selects the best justified action, observes the outcome, and consolidates lessons into future decision behavior.

13. Bibliography

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Appendix A: MVP Backlog Snapshot

Epic	P0 Output
Core Engine	FastAPI skeleton, Pydantic schemas, run_taco_loop orchestrator.
Unknown Data	Known facts, missing facts, contradictions, assumptions, stale items.
Environment	Domain, time pressure, volatility, adversarial pressure, tool state.
Uncertainty	Total uncertainty score and loop-speed recommendation.
Decision	Candidate generation, utility scoring, selected and rejected actions.
Guardrails	Rule, permission, scope, safety, reversibility checks.
Audit	Hash-chained decision trace and export.
Benchmark	Baseline vs TACO scenario runner and score report.
Dashboard	Scenario runner, decision card, unknowns panel, environment panel, audit timeline.
Operator Queue	Escalation, approve, deny, request-more-data, audit note.